

Column-by-column isometry synthesis (Iten 1501.06911) with Berry Eq. 24 QROAM layers

$n = [3, 4, 5, 6, 7, 8]$ ($N = 2^n$), phase precision $b = 32$, $n_{\text{blocks}} = 1$.

All curves use the Toffoli-optimal QROAM block size (optimal_T=True).

Toffoli = get_Toffoli_counts ($n_{\text{ccz-equivalent}}$); qubits = QubitCount peak logical.

Methods:

- Column-by-column: K column operations, each a staircase of merged multiplexed single-qubit gates ($\Delta_s \cdot C_{n-1-s}^u(U_s)$) realized via Eq. 24 ($R(\phi) H R(\theta) H + \text{QROAM phases}$) plus subleading multi-controlled gates. Leading cost $\sim K \cdot 2^n / \Lambda$; synthesizes only the K needed columns.
- Reflection (LKS): K Householder reflections (arXiv:1812.00954 Sec. 4).
- Interferometer: full-unitary rectangular mesh (Berry Sec. III A); always N columns.

Block-aware: BlockIsometryColumnSynthesisQROAM carries an optional read-only block register ($\sum_j |j\rangle\langle j| (x) V_j$); shown here with $n_{\text{blocks}}=1$ (single isometry).

Fitted power laws $y = c \cdot N^\alpha$:

Column-by-column, $K=N$: T $\alpha=1.426$ $c=1.65e+02$; Q $\alpha=0.121$ $c=7.14e+01$

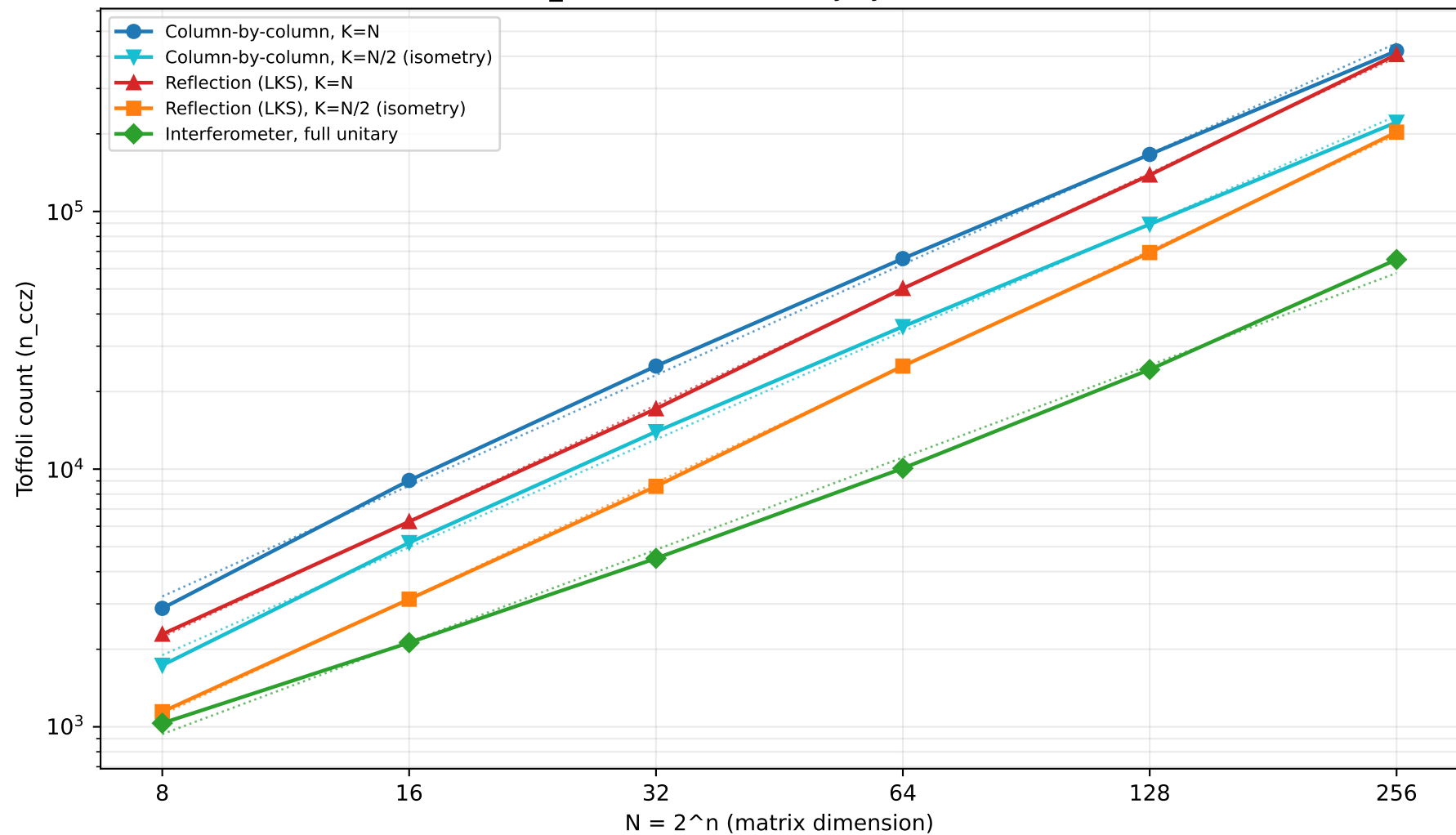
Column-by-column, $K=N/2$ (isometry): T $\alpha=1.391$ $c=1.05e+02$; Q $\alpha=0.121$ $c=7.14e+01$

Reflection (LKS), $K=N$: T $\alpha=1.495$ $c=9.97e+01$; Q $\alpha=0.244$ $c=3.87e+01$

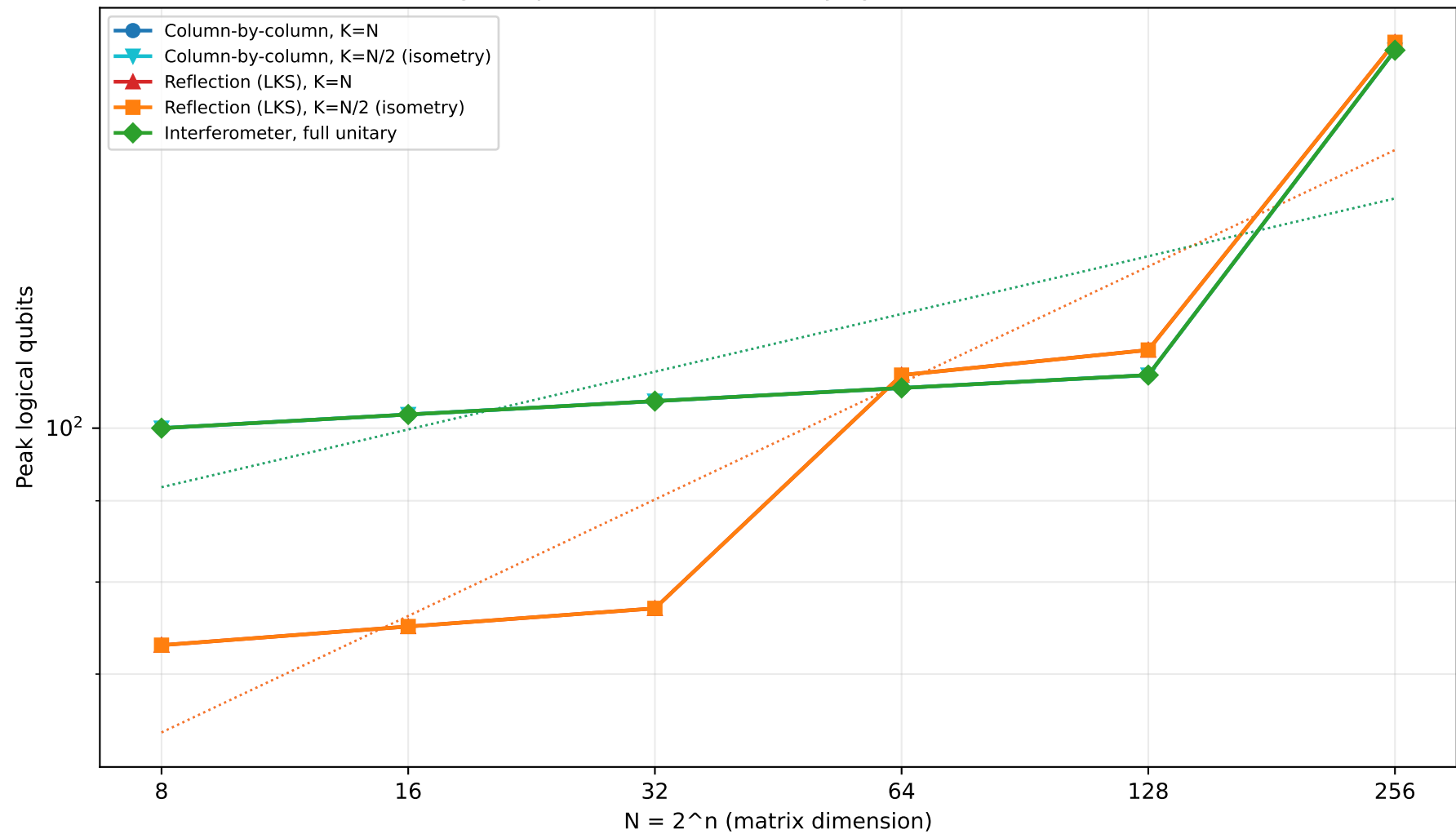
Reflection (LKS), $K=N/2$ (isometry): T $\alpha=1.495$ $c=4.99e+01$; Q $\alpha=0.244$ $c=3.87e+01$

Interferometer, full unitary: T $\alpha=1.189$ $c=7.90e+01$; Q $\alpha=0.121$ $c=7.14e+01$

Toffoli count (n_{ccz}) vs N for isometry synthesis methods ($b=32$)



Peak logical qubits vs N for isometry synthesis methods (b=32)



Toffoli (n_ccz) counts by method (Toffoli-optimal QROAM)

n	N	Column-by-column, K=N	Column-by-column, K=N/2 (isomorphism)	Reflection (LKS), K=N/2	Reflection (LKS), K=N/2 (isomorphism)	Interferometer, full unitary
3	8	2,882	1,733	2,288	1,144	1,033
4	16	9,039	5,185	6,256	3,128	2,121
5	32	25,120	13,977	17,152	8,576	4,505
6	64	65,617	35,737	50,240	25,120	10,073
7	128	166,418	89,177	138,496	69,248	24,345
8	256	419,828	221,882	406,272	203,136	65,082